# Remote Sensing Fine-Grained Ship Data Augmentation Pipeline With Local-Aware Progressive Image-to-Image Translation

Bo Liu, Ligang Li, Qi Xiao, Wei Ni, and Zhen Yang

***Abstract*— Remote sensing image ship fine-grained classification is a challenging vision problem due to factors, such as interclass similarity, real-world image scarcity, and class imbalance. Data augmentation aims to solve these problems from the data perspective. Most of these methods cannot work effectively in complex scenes, which restricts their practical application. Here, we propose a novel data augmentation pipeline based on local-aware image translation to achieve the representation mapping between cross-domain corresponding instances. Our pipeline contains three modules: imaging simulation system (ISS), local-aware progressive image-to-image translation (LoPIT), and image harmonization (IH) modules. The ISS module generates simulated images with correct appearance and diverse features based on the input requirement information. To tackle the domain gap between simulated image and real-world image, we propose the local-aware CycleGAN in the LoPIT module to achieve mapping based on local-aware learning and apply two submodules to progressively complete the remote sensing image global cross-domain translation. The IH module uses IH technology to coordinate the visual appearance between the foreground and background to generate sufficient remote sensing ship images with precise representation, photorealistic style, and harmonious features. Moreover, we present a mixed dataset including real-world images and our synthetic images for remote sensing image fine-grained ship classification. Our dataset named RSSA-12 contains 12 categories of ship targets in 3831 images with high-quality annotated category labels, effectively alleviating the long-tail problem of existing datasets. Experimental results demonstrate that our progressive pipeline outperforms the state-of-the-art data augmentation method on the remote sensing fine-grained ship classification task.**

***Index Terms*— Fine-grained ship classification, image harmonization (IH), image-to-image (I2I) translation, imaging simulation, remote sensing.**

## I. Introduction

DEEP learning (DL) models have made incredible progress in numerous tasks. This has been fueled by the high-capacity network architectures, the booming calculational power, and the availability of large-scale dataset. Deep neural networks (DNNs) have been successfully applied to computer vision tasks, such as image classification [1], [2], [3], [4], object detection [5], [6], [7], and image segmentation [8], [9], [10], [11], due to the development of convolution neural networks (CNNs).

With the rapid development of remote sensing equipment and technologies, applications based on remote sensing images emerge endlessly, including remote sensing scene classification [12], [13], [14], [15], geographical image data analysis [16], [17], and geographic semantic segmentation [18], [19]. There is unanimous agreement that the accuracy of feature expression described by the deep network model depends on large-scale available data. The availability of remote sensing images is not only limited to the instability of the imaging system but affected by severe environments, such as extreme weather conditions, illumination, atmosphere, and cloud occlusion [20]. For example, in the field of remote sensing image fine-grained ship classification, real-world datasets are confronted with many problems, such as poor availability and tedious annotations. When the training data are insufficient, the trained network model cannot completely align the distribution characteristics of data unilaterally, and the generalizability of the model is limited.

Researchers have proposed a series of techniques and methods to enhance robustness from the perspective of network architecture [21], [22] and data. Data augmentation represents a more comprehensive set of possible data, thus minimizing the gap between the training and validation set, as well as any future testing sets. These traditional augmentation methods, which are easy to implement and widely used, put forward requirements for the scale and diversity of original data that cannot be ignored. When the training data are extremely scarce, the effectiveness of these methods is limited. Generative adversarial network (GAN) [23] has attracted the attention of researchers because of its remarkable feature description and expression ability. However, there will be some thorny difficulties in the training process, such as poor stability, gradient disappearance, and pattern collapse. Synthetic data have the advantages of easy access and convenient annotation. Unfortunately, most synthetic data are not photorealistic or akin to the real-world scene. The network model relies excessively on synthetic data, which will weaken its capability in the real-world domain. This dilemma is regarded as a domain divergence phenomenon, which can be alleviated by

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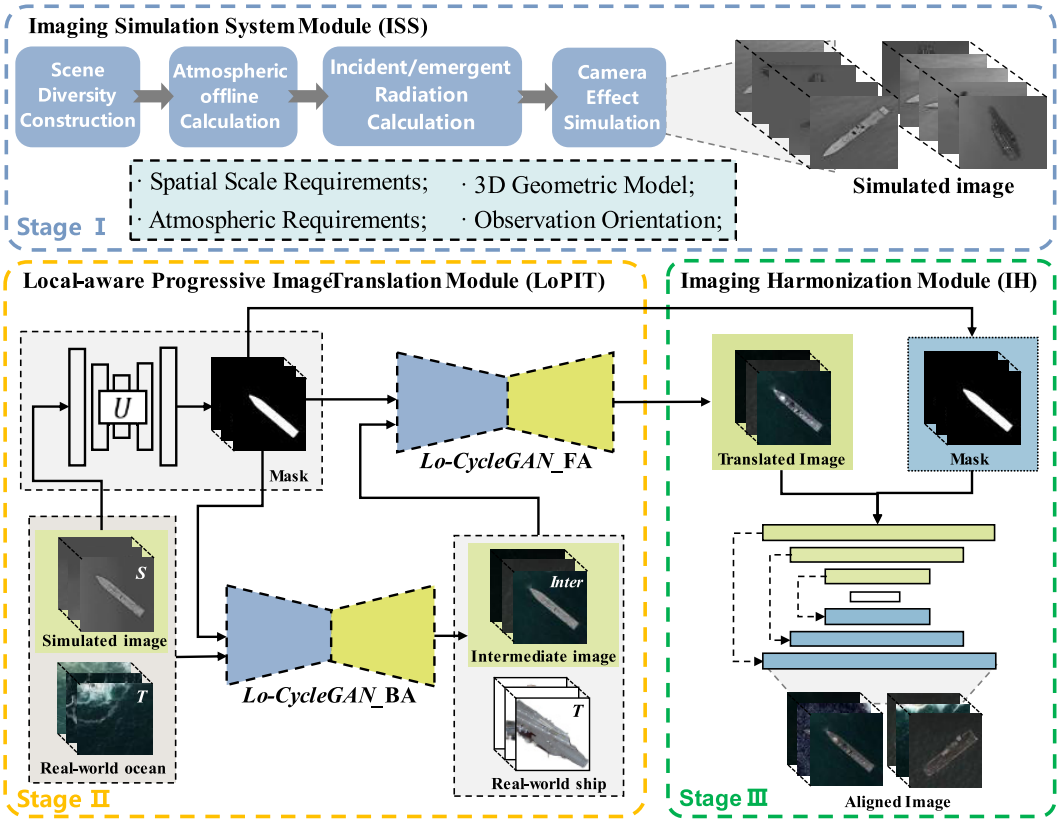



Fig. 1. Framework of our progressive data augmentation pipeline.

image-to-image (I2I) translation methods. Additionally, paired corresponding remote sensing images in different domains are costly, and the unsupervised image-to-image translation (UNIT) [24], [25] has extensive serviceability. Nevertheless, there are great differences between the foreground and background of remote sensing ship images, and the global-aware UNIT will negatively interfere with the precise expression of the instance features. Due to the aforementioned limitations, local-aware I2I network architecture and photorealistic remote sensing ship synthetic images are desirable.

To solve these problems, we propose a novel data augmentation pipeline (see Fig. 1). First, for the ship target with extremely scarce real-world images, we established a visible light imaging simulation system (ISS), completed the ocean scene construction and ship target modeling, and generated six classes ship simulated images. Nevertheless, the obvious visual style distance exists in the simulation and real-world domain. We propose the local-aware CycleGAN (Lo-CycleGAN) to complete the image translation of the background and foreground, respectively. Considering the gap in brightness and color between the foreground and background of the translated image, we apply image harmonization (IH) technology to adjust the foreground in the translated image to make it harmonious with the background. We developed sufficient experiments on the remote sensing image fine-grained ship classification task. The extensive experimental results demonstrate the effectiveness of our proposed pipeline. The main contributions of this article are summarized as follows.

1) We propose a novel data augmentation pipeline based on local-aware image translation to achieve the same property corresponding regions cross-domain translation. Based on this pipeline, we generate sufficient remote sensing ship images with precise representation, photorealistic style, and harmonious features.

2) We use IH technology to adjust the foreground of the translated image. To the best of our knowledge, we are the first to apply IH technology to the data augmentation of remote sensing image fine-grained ship classification task.

3) We present a mixed dataset (named RSSA-12) for remote sensing image fine-grained ship classification. The mixed dataset contains 3831 images with 12 category labels, of which six categories include the images generated by our pipeline and real-world images and the other six categories only include real-world images. Extensive experiments are conducted on our dataset to verify the effectiveness of our pipeline.

The rest of this article is organized as follows. Section II introduces the related work of this study. Section III describes the mathematical expression and principle procedure of our pipeline. Section IV provides the experiment implementation details and dataset statistics. Section V presents the experimental results and analyzes the performance. Finally, Section VI concludes this article.

## II. Related Work

### A. Data Augmentation

*1) Traditional Image Data Augmentation Methods:* Traditional image data augmentation methods rely on image

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processing technology [26] to achieve data augmentation and image quality optimization. They are divided into three categories: geometric transformation, color transformation, and pixel transformation. Geometric transformation refers to the geometric transformation of the image, including flipping, rotation, clipping, scaling, and so on. The color transformation will change the image content. Common operations include noise, blur, color channel transformation, erasing, filling, random erasing, and so on. Pixel transformation is a data augmentation method that operates on image pixels. In the application scenario, the limited applicability is the disadvantage of traditional image data augmentation methods.

*2) Data Augmentation Method Based on DL:* With the rapid development and wide application of DL, researchers have applied DL technology to data augmentation and achieved remarkable scientific research results. Data augmentation based on GAN is the mainstream method at this stage. The idea of GAN [23] application in data augmentation task is to generate new training data and to expand model training samples. Researchers design different network architectures and loss functions to promote the performance of the GAN series. The discriminator and generator of DCGAN [27] employ CNN to replace the multilayer perceptron in GANs. CycleGAN [25] is a major model in the field of image translation, which achieves image translation with unpaired data. Conditional GAN [28] modifies the algorithm architecture through conditional constraints and optimizes the data generation process.

*3) Synthetic Data-Based Data Augmentation Method:* The innovation of imaging simulation and virtual engines provides technical and software support for constructing synthetic datasets. In recent years, researchers have attempted to use synthetic images to partially replace real-world images to train the model and improve the robustness of the model in few-shot learning [29], [30]. Unfortunately, synthetic data cannot perfectly align with real-world features. Completely relying on synthetic data to train the model may cause low performance in practical scenarios [31]. Sim2RealNet [32] is a newly designed network based on neural style transfer (NST) that is employed to complete style transfer from simulated samples to real-world images. NST is not enough to improve the appearance representation of the image, which requires more calculation quantity and running time.

### B. I2I Translation

I2I translation aims to learn the mapping between images in two visual domains. Its applications include image editing [33], image-colorization [34], and style transfer [35]. Paired I2I translation completes the mapping of the input image from the source domain to the target domain by the adversarial loss [23]. Pix2pix [36] proposes a unified framework based on conditional GAN [28] to achieve "pixel-level style transfer." In general, paired remote sensing images belonging to two domains are demanding and costly, so unpaired I2I translation has more practical significance. CycleGAN [25] limits the generator to retain the source domain image features by adding the cycle-consistency instead of the constraints used in conditional GAN, breaking through the limitation of Pix2pix on paired data. UNIT and MUNIT [37] assume that diverse data spaces share the "content" latent space and transform the problem into solving a hidden space. DRIT series [38] and CUT [39] are based on the framework of contrastive learning to maximize the mutual information between the input image and the corresponding patch in the target domain, so that it focuses on the content rather than the style of the object in the image. However, these methods focus on the image-level feature information and are committed to map the entire expression of the source domain to the target domain. These models based on the global-level feature learning cannot produce more detailed texture characters in the complex structure image translation. Unlike the above approaches, we use binary images to distinguish translation regions and invariant regions and complete the same property corresponding regions cross-domain translation. The model based on the same property local aware can avoid the image distortion caused by feature mismatch and improve the applicability of image translation technology in complex scenes.

### C. Image Harmonization

Traditional IH methods focus on matching low-level appearance statistics, such as color statistics [40], gradient information [41], multiscale statistical features [42], and semantic information [43]. These methods match appearance to harmonize a composite image while paying less attention to visual realism. In recent years, with the development of DL technology, more DL-based methods learn a translation from composite images to real-world images, with domain verification discriminator [44], background-guided domain translation [45], and background-to-foreground style transfer [46]. Due to the complexity of remote sensing scenes and the lack of relevant paired datasets, IH technology is rarely applied to the fine-grained ship classification field in remote sensing images before our work.

## III. Proposed Method

The scarcity and low-quality of real-world images are the key causes restricting remote sensing image fine-grained ship classification. The advantage of imaging simulation technology is to generate sufficient simulated images and alleviate the poor performance caused by inadequate real-world data. However, the feature distribution of simulated samples and real-world images cannot be perfectly aligned. I2I translation can restrict the distribution of generated samples and input images as consistent as possible. Due to the complex details of remote sensing images and the apparent divergence in feature representation between different instances, the applicability of I2I translation based on image-level feature learning in remote sensing scenes is insufficient. To alleviate the above problems, we propose a data augmentation pipeline based on local-aware progressive image-to-image translation (LoPIT).

### A. Overall Architecture

As Fig. 1 shows, the proposed data augmentation pipeline included three critical stages, which were achieved by three

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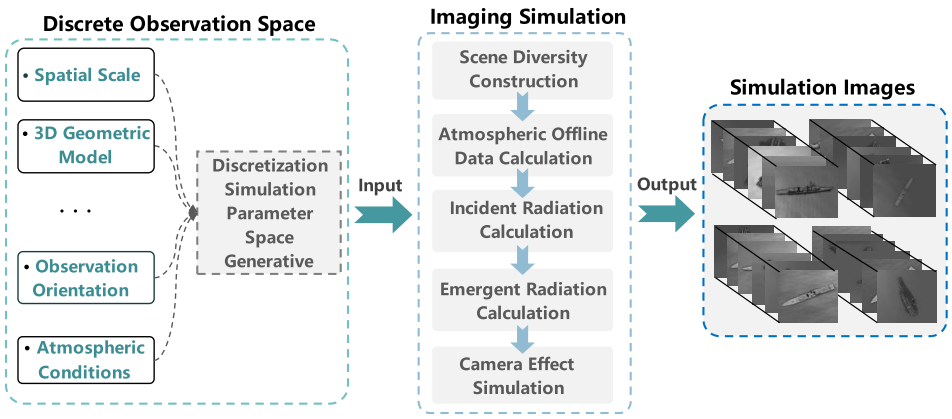



Fig. 2. Framework of our visible light ISS. First, we select the imaging parameters that affect the transmission characteristics of visible light, such as spatial scale, observation azimuth, and atmospheric conditions. Then, we organize these imaging parameters into discrete observation space (left) to meet the input requirements of imaging simulation (middle). Finally, we conduct sample simulation combined with the visible light imaging simulation and generate simulated images with realistic texture details (right).

modules: ISS module, LoPIT module, and IH module. The ISS module generated simulated images based on the input requirement information, such as the spatial scale and observation parameters, which were used as the characteristic diversity data source of our pipeline. To avoid the disadvantage of global perception image feature distribution, based on the idea of local perception and progressive translation, we use two submodules in the LoPIT module, each containing an Lo-CycleGAN model, to achieve global-level cross-domain translation. Then, we applied IH technology to adjust the foreground in the translated image, make it coordinate with the background, and make the overall expression of the image closer to the real world. Finally, we generated sufficient remote sensing ship images with precise representation, photorealistic style, and harmonious features. Thus, our progressive data augmentation pipeline addressed data scarcity for remote sensing image fine-grained ship classification tasks from the perspective of both imaging simulation and LoPIT. We will represent the mathematical support and implementation details of our proposed data augmentation pipeline in Sections III-B–III-D.

### *B. ISS Module*

We considered the geometric and spectral characteristics of the ship target and ocean background and the spatiotemporal relationship of imaging detection. Then, according to the radiation transfer theory, we implemented the radiation imaging simulation modeling calculation including ship target and ocean background. Based on clarifying the visible light imaging radiation process and transmission mechanism, we organized the imaging parameters affecting the radiation transmission characteristics (such as spatial scale, observation orientation, atmospheric conditions, etc.) into discrete observation space to meet the basic requirements of imaging simulation input.

These parameters were regarded as the input of the visible light ISS, and the remote sensing ship simulated image was generated according to the basic optical imaging principle. Fig. 2 shows the framework of our ISS. Combined with the geometric dimensions of ship targets, we constructed the three-dimensional (3-D) geometric models of six classes ships with 3-Ds Max modeling software. To enrich the feature diversity of the generated simulated image, we set different combinations of observation parameters (including atmospheric visibility, observation distance, and observation angle) for ship targets. Fig. 3 shows the generated partial simulated images of various ships.

Fig. 3 lists partial simulated images of various ships generated by the ISS module. In the process of radiation calculation, the calculation parameters of MODTRAN were set as follows: visibility distance 23 km, solar altitude angle 15°, midlatitude summer atmospheric model, and marine aerosol type. The parameter combination of external observation conditions included observation distance and observation angle. The observation distance referred to the distance between the ship target and the camera platform. The observation angle was determined by the observation altitude angle and azimuth angle. The observation altitude angle was the included angle between the camera optical axis and the perpendicular normal vector of the sea level. The observation azimuth was the included angle between the sea-level projection and the bow direction (clockwise was positive and counterclockwise was negative). Table I lists the observation parameters of these simulated images in Fig. 3.

### *C. LoPIT Module*

Image translation based on image-level feature learning could not establish the mapping relationship between corresponding cross-domain instances, resulting in the feature's distortion of the generated image. Achieving UNIT between two distant domains was challenging; hence, we proposed an intermediate domain to make the image translation task easier, instead of directly narrowing the feature distance between the source and target domains. Local perception and progressive translation are the fundamental strategies of the LoPIT module. This module is composed of two submodules named background alignment (BA) and foreground alignment (FA) connected in sequence. Each submodule contains a Lo-CycleGAN to complete the corresponding assignments.

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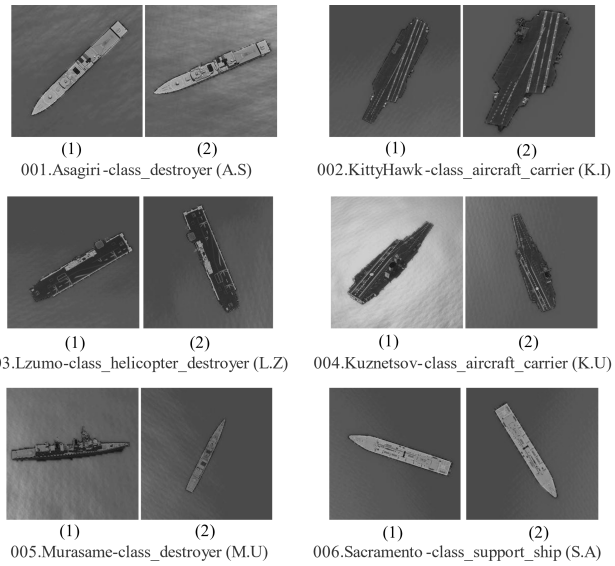



Fig. 3. Partial simulated images of various ships were generated by our ISS. There are six classes of ship in these simulated images, namely, Asagiri-class_destroyer (A.S), Kittyhawk-class_aircraft_carrier (K.I), Lzumo-class_helicopter_destroyer (L.Z), Kuznetsov-class_aircraft_carrier (K.U), Murasame-class_destroyer (M.U), and Sacramento-class_support_ship (S.A).

TABLE I
IMAGING SIMULATION PARAMETER SETTINGS ABOUT FIG. 3

| Types | No. | Imaging Simulation System Parameter Settings | | |
|---|---|---|---|---|
| | | Observation Distance (km) | Observation Azimuth (°) | Observation Altitude Angle (°) |
| A.S | (1) | 1.5 | 45.5 | 90.0 |
| | (2) | 1.5 | 68.2 | 80.0 |
| K.I | (1) | 2.0 | 31.2 | 80.0 |
| | (2) | 1.5 | 31.2 | 80.0 |
| L.Z | (1) | 2.0 | 55.0 | 90.0 |
| | (2) | 2.0 | -25.8 | 90.0 |
| K.U | (1) | 2.0 | -141.5 | 70.0 |
| | (2) | 2.0 | 166.7 | 90.0 |
| M.U | (1) | 2.0 | -91.6 | 70.0 |
| | (2) | 4.0 | -152.4 | 90.0 |
| S.A. | (1) | 1.5 | 114.9 | 90.0 |
| | (2) | 1.5 | -31.2 | 90.0 |

Next, we will introduce this module in detail in the order of the basic model (Lo-CycleGAN), BA submodule, and FA submodule. We denoted the *source*, *intermediate*, and *target* domains as $S$, Inter, and $T$, respectively. Here, the source domain referred to the simulation domain and the target domain referred to the real-world domain. As Fig. 4 shows, we performed image translation to the real-world domain bridged by the intermediate domain.

*1) Local-Aware CycleGAN:* To avoid the one-sidedness of global aware, we proposed Lo-CycleGAN to complete precise local aware from two perspectives, achieve instance-level feature mapping, and improve the application feasibility of image translation technology in the challenging remote sensing ship fine-grained classification scene.

As Fig. 5(a) shows, the input of Lo-CycleGAN had three parts: source domain image $I_s \sim P_{\text{source}}(I_s)$, $P_{\text{source}}(I_s)$ referred to the characteristic distribution of source domain; the binary mask $M$ indicated the translation region $R_{s_t}$ and invariant region $R_{s_i}$ in the source domain image; and

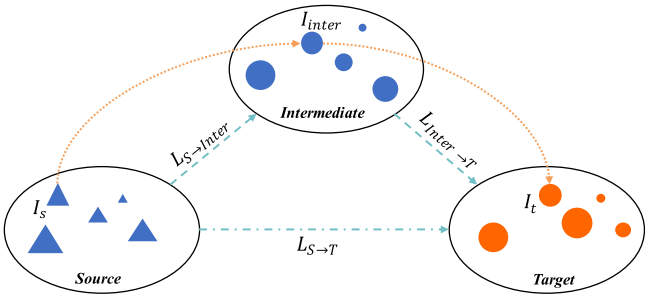



Fig. 4. Illustration of our LoPIT. The distance $L_{S\to T}$ between source (simulation) domain $S$ and target (real world) domain $T$ cannot be neglected. The traditional image translation methods based on image-level feature learning will be disturbed by different regional features, resulting in feature doping and chaotic expression, and the effect of image translation is not ideal. We propose the LoPIT module, which introduces the intermediate domain Inter to bridge source domain and target domain. In other words, the whole I2I translation process has changed from $S \to T$ to $S \to \text{Inter} \to T$.

target domain image $I_t \sim P_{\text{target}}(I_t)$, $P_{\text{target}}(I_t)$ referred to the characteristic distribution of target domain.

The input source domain image $I_s$ was regarded as composed of translation region $R_{s_t}$ and invariant regions $R_{s_i}$, i.e., $I_s = R_{s_t} + R_{s_i}$. The translation region $R_{s_t}$ and invariant region $R_{s_i}$ were aligned with the characteristic distribution of the source domain $P_{\text{source}}(I_s)$, i.e., $R_{s_t} \sim P_{\text{source}}(I_s)$ and $R_{s_i} \sim P_{\text{source}}(I_s)$. The principal function of Lo-CycleGAN is to perceive the feature distribution of the translation region and achieve the region mapping operation from source domain to target domain. This process was expressed as

$$R_{o_t} + R_{o_i} = I_{\text{out}} = F_{\text{Lo-CycleGAN}}(I_s, M, I_t) \tag{1}$$

where $F_{\text{Lo-CycleGAN}}(\cdot)$ referred to the Lo-CycleGAN model, and $I_{\text{out}}$ was the output image, which was composed of the translation region $R_{o_t}$ and invariant region $R_{o_i}$

$$R_{o_t} \sim \widetilde{P_{\text{target}}(I_t)}, \quad R_{o_i} = R_{s_i} \sim P_{\text{source}}(I_s). \tag{2}$$

Lo-CycleGAN consists of two generators ($G_{S\to T}, G_{T\to S}$) and two discriminators ($D_S, D_T$), in which $G_{S\to T}$ develops the mapping from source domain to target domain ($S \to T$), $G_{T\to S}$ develops the mapping from target domain to source domain ($T \to S$), $D_S$ discriminates whether the image belongs to source domain, and $D_T$ discriminates whether the image belongs to target domain.

Theoretically, when the model was trained to converge, the distribution of the translation region in the output image $\widetilde{P_{\text{target}}(I_t)}$ was aligned with the target domain $P_{\text{target}}(I_t)$ as much as possible, i.e., $\widetilde{P_{\text{target}}(I_t)} = P_{\text{target}}(I_t)$. At this time, the target domain discriminator $D_T$ could not identify effectively, i.e., $D_T(G_{S\to T}(I_s, M)) = (1/2)$.

In the image translation process, we extracted and retained the invariant region $R_{s_i}$ with the binary mask $M$. After completing the feature mapping of the translation region, we covered the invariant region to the corresponding position of the generated image, i.e., $R_{s_i} = R_{o_i}$. In this way, we ignored any possible influence of image translation operation on the invariant region $R_{s_i}$, avoided the interference of irrelevant features, and ensured the relative independence

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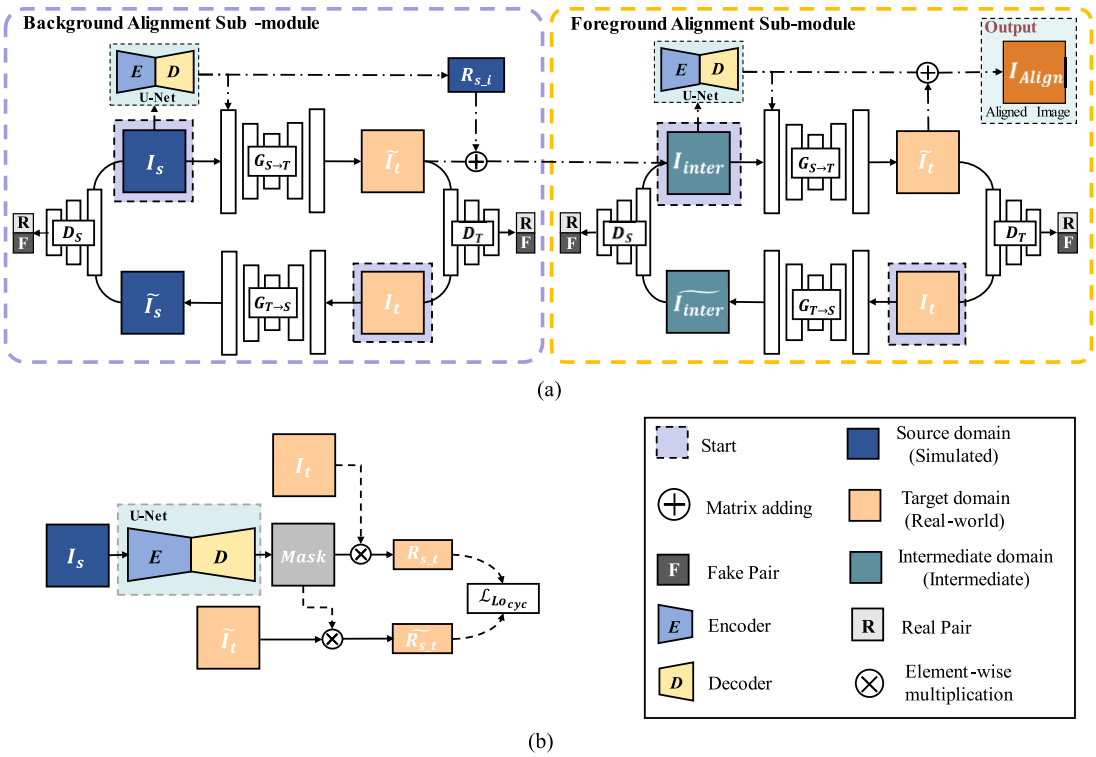



Fig. 5. Detailed structure and procedure principle diagram of our LoPIT module. (a) LoPIT module includes two submodules, BA submodule and FA submodule. (b) Illustration of the $L_{\text{Lo}_{\text{cyc}}}$ in BA submodule.

of the invariant region. The construction of loss function affected the model's convergence. We ignored the mapping error of invariant regions, calculated the cycle-consistency loss of translation regions, and then completed the local aware. Inspired by this idea, we modified the loss function of the image translation model and proposed the local-aware cycle-consistency loss. In the process of iterative training, the model was prompted to match the feature representation of the translation region $R_{s_t}$ and then interfere with the model's convergence direction.

Ideally, for each image $I_s$ from source domain $S$, the image translation cycle was recurring, and the process was described as $I_s \rightarrow G_{S\rightarrow T}(I_s) \rightarrow G_{T\rightarrow S}(G_{S\rightarrow T}(I_s)) \approx I_s$. Similarly, for each target domain image $I_t$, two mappings had to achieve reverse consistency, i.e., $I_t \rightarrow G_{T\rightarrow S}(I_t) \rightarrow G_{S\rightarrow T}(G_{T\rightarrow S}(I_t)) \approx I_t$. The consistency of the above two directions was called forward/backward cycle consistency. In the local-aware I2I translation process, our goal was to achieve the mapping of the translation region $R_{s_t}$ and retain the invariant region $R_{s_i}$. To clarify this purpose, we neglected the cycle-consistency loss of the invariant region $R_{s_i}$ in the translation direction $S \rightarrow T$ and proposed the local-aware cycle-consistency loss $\mathcal{L}_{\text{Lo}_{\text{cyc}}}$. We described $\mathcal{L}_{\text{Lo}_{\text{cyc}}}$ as

$$\begin{aligned}&\mathcal{L}_{\text{Lo}_{\text{cyc}}}(G_{S\rightarrow T}, G_{T\rightarrow S})\\&\quad=\mathbb{E}_{I_s\sim P_{\text{source}}(I_s)}\left[\left\|G_{T\rightarrow S}(G_{S\rightarrow T}(I_s))\circ M - I_s\circ M\right\|_1\right]\\&\qquad+\mathbb{E}_{I_t\sim P_{\text{target}}(I_t)}\left[\left\|G_{S\rightarrow T}(G_{T\rightarrow S}(I_t)) - I_t\right\|_1\right]\end{aligned}\tag{3}$$

where $\circ$ is the Hadamard product. In general, there are different mapping paths from the source domain $S$ to the target domain $T$. The local-aware cycle-consistency loss $\mathcal{L}_{\text{Lo}_{\text{cyc}}}$ reduces the possibility of mapping directions, and the simplest way to achieve mapping is to maintain the structure characters of the translation region $R_{s_t}$.

Adversarial training is a major pattern to strengthen the robustness of networks. Adversarial learning can train mapping procedures $S \rightarrow T$ and $T \rightarrow S$. These two mappings produced output images that matched the feature distribution of the corresponding source domain $S$ and target domain $T$, respectively.

We refer to the setting of [25], for the translation direct $S \rightarrow T$, the adversarial loss was expressed as

$$\begin{aligned}&\mathcal{L}_{\text{GAN}}(G_{S\rightarrow T}, D_T, S, T)\\&\quad=\mathbb{E}_{I_t\sim P_{\text{target}}(I_t)}\left[\log D_T(I_t)\right]\\&\qquad+\mathbb{E}_{I_s\sim P_{\text{source}}(I_s)}\left[\log(1 - D_T(G_{S\rightarrow T}(I_s)))\right]\end{aligned}\tag{4}$$

where $\log D_T(I_t)$ denoted the probability that the discriminator $D_T$ determined the target domain image $I_t$ as a target domain image. $G_{S\rightarrow T}(I_s)$ represented the output image of the generator $G_{S\rightarrow T}$. Note that $G_{S\rightarrow T}(I_s)$ was not covered by the reserved invariance region $R_{s_i}$. $\log(1 - D_T(G_{S\rightarrow T}(I_s)))$ denoted the probability that the discriminator $D_T$ determined $G_{S\rightarrow T}(I_s)$ as a fake target domain image. Similarly, for another translation direct generator $G_{T\rightarrow S}$ and corresponding discriminator $D_S$, the adversarial loss was described as $\mathcal{L}_{\text{GAN}}(G_{T\rightarrow S}, D_S, S, T)$.

For the generator $G_{S\rightarrow T}$, if we input $I_t$ into the generator $G_{S\rightarrow T}$, ideally, there should be $G_{S\rightarrow T}(I_t) = I_t$. This operation could verify that the generator $G_{S\rightarrow T}$ had the translation capability from source domain $S$ to target domain $T$. Therefore, $G_{S\rightarrow T}(I_t)$ and $I_t$ had to be as identical as possible.

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This process was optimized by adding identity loss $\mathcal{L}_{\text{idt}}$, which was defined as

$$\mathcal{L}_{\text{idt}}(G_{S\to T}, G_{T\to S}) = \mathbb{E}_{I_s\sim P_{\text{source}}(I_s)}\left[\|G_{T\to S}(I_s) - I_s\|_1\right] + \mathbb{E}_{I_t\sim P_{\text{target}}(I_t)}\left[\|G_{S\to T}(I_t) - I_t\|_1\right]. \quad (5)$$

Our full objective was

$$\mathcal{L}(G_{S\to T}, G_{T\to s}, D_S, D_T) = \mathcal{L}_{\text{GAN}}(G_{S\to T}, D_T, X, Y) + \mathcal{L}_{\text{GAN}}(G_{T\to S}, D_S, Y, X) + \lambda_{\text{Lo}_{\text{cyc}}}\mathcal{L}_{\text{Lo}_{\text{cyc}}} + \lambda_{\text{idt}}\mathcal{L}_{\text{idt}} \quad (6)$$

where $\lambda_{\text{Lo}_{\text{cyc}}}$ and $\lambda_{\text{idt}}$ were the weight factor of the $\mathcal{L}_{\text{Lo}_{\text{cyc}}}(G_{S\to T}, G_{T\to S})$ and $\mathcal{L}_{\text{idt}}(G_{S\to T}, G_{T\to S})$, respectively. Our overall min–max loss function was integrally defined as

$$G^*_{S\to T}, G^*_{T\to S} = \arg\min_{G_{S\to T}, G_{T\to S}} \max_{D_S, D_T} \mathcal{L}(G_{S\to T}, G_{T\to s}, D_S, D_T). \quad (7)$$

The above is the principle of local-aware image translation. In the field of remote sensing ship image fine-grained classification, the translation of the overall image is achieved through two submodules, and each submodule completes the feature alignment of the translation region $R_{s_t}$ with an Lo-CycleGAN.

*2) BA Submodule:* In the BA process, the remote sensing simulated images composed the source domain, and the real-world ocean images constituted the target domain. The translation region $R_{s_t}$ was the ocean background in the simulated image, and the invariant region $R_{s_i}$ was the ship target.

Fig. 5 shows the principle framework of our BA submodule. The input of this submodule included the ocean background image from the real-world domain, the remote sensing ship simulated image from the simulation domain, and the corresponding background mask indicating the ocean background (translation region). It was necessary to ensure the representation invariance of the ship target (invariant region) throughout the BA process. To explicit the direction of the translation, we neglected the foreground and calculated the cycle-consistency loss of the background region. The schematic of $\mathcal{L}_{\text{Lo}_{\text{cyc}}}$ in the BA submodule is shown in Fig. 5(b).

The BA submodule's assignment was to align the representation distribution of the simulated image with the real-world domain, while retaining the original texture and structural features. The characteristics of ocean environment in different regions might be diverse, such as surface color, wave shape, ripple texture, and other significant visual content features. To expand the diversity of marine background style representation, we collected over 3300 remote sensing ocean images from several regions on Google Earth. These remote sensing ocean scene images ensured the diversity of sea surface style features.

The output image of the BA submodule was called the intermediate image. Its ocean background feature distribution was close to the real-world domain, and the ship target characteristics still matched the simulation domain. The global feature distribution of intermediate image was called intermediate domain. Fundamentally, we regarded the intermediate domain as the bridge between the source domain and the target domain to reduce the challenge of cross-domain translation.



Fig. 6. Illustration of partial real-world ships images for FA. We apply image segmentation technology to extract the foreground (ship target) in the real-world remote sensing image to avoid the interference of other regional representations (ocean, port background, etc.).

*3) FA Submodule:* In the FA process, the intermediate images were regarded as the source domain, and the real-world ship images constituted the target domain. The translation region $R_{s_t}$ was the ship target, and the invariant region $R_{s_i}$ was the ocean background in the intermediate image. The FA submodule's assignment was to align the representation distribution of the intermediate image with the real-world domain through real-world ship images, while retaining the original texture and structural features of these specific ships.

Fig. 5(a) shows the principle framework of our FA submodule. The inputs of this submodule included the ship target images from the real-world domain, the intermediate image from the intermediate domain, and the corresponding foreground mask indicating the ship target (translation region). Fig. 6 shows the various ship targets in real-world remote sensing images. In this process, we neglected the ocean background and calculated the cycle-consistency loss of the ship target. To distinguish the images of different processing stages, we named the images processed by BA and FA submodules as translated images.

### D. IH Module

Since the above BA and FA procedures were like image synthesis operations, these processes did not explicitly establish the correlation between foreground and background. In this module, we applied IH technology to adjust the feature representation of the ship foreground, made it consistent with the style expression of the background, and improved the image's overall authenticity. The generator network architecture of IH module is shown in Fig. 7.

Given an intermediate image (composite image) $I_{\text{inter}}$ and the corresponding foreground binary mask $M$, the task of IH module was to use a generator $G_h$ to construct $I_{\text{inter}}$ with a harmonized appearance, i.e., $G_h(I_{\text{inter}}, M)$. The IH module was a learnable model that was as close to the real-world training image $I$ by a reconstruction loss

$$\mathcal{L}_{\text{rec}}(G_h, I, I_{\text{inter}}, M) = \|G_h(I_{\text{inter}}, M) - I\|_1. \quad (8)$$

Inspired by the GAN [23], the IH model also uses the strategy of adversarial training. Specifically, the adversarial loss in this

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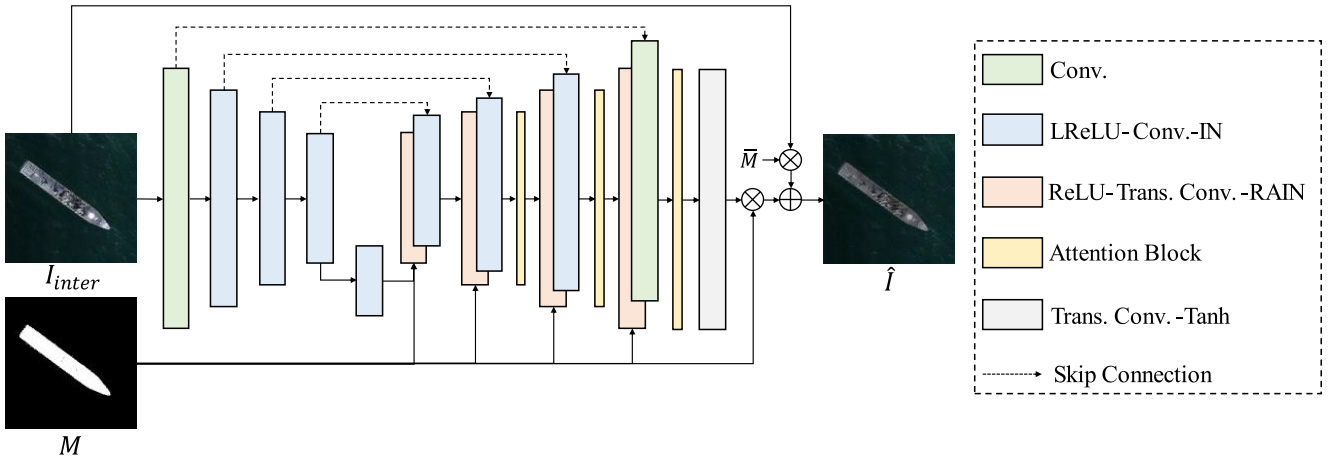



Fig. 7. Detailed structure and procedure principle diagram of our IH generator. The meaning of several abbreviations in the figure is: Conv. = Convolution, Trans. = Transposed, and RAIN = RAIN unit. The detailed structure and principles of the RAIN unit in the generator are consistent with [23].

module can be expressed as

$$\mathcal{L}_{\text{adv}}\left(D_h, I, \hat{I}\right) = \mathbb{E}_I[\max(0, 1 - D_h(I))] + \mathbb{E}_{\hat{I}}\left[\max\left(0, 1 + D_h\left(\hat{I}\right)\right)\right] \quad (9)$$

and

$$\mathcal{L}_{\text{adv}}(G_h, I_{\text{inter}}, M) = -\mathbb{E}_{I_{\text{inter}}}[D_h(G_h(I_{\text{inter}}, M))] \quad (10)$$

where global discriminator $D_h$ tries to distinguish between real-world training image $I$ and harmonized samples $\hat{I}$. The generator $G_h$ aims to generate images that look similar to the real-world appearance. Introducing adversarial loss can learn the generator model $G_h$ that generate images as photorealistic as the real world.

IH operation can be regarded as a domain translation process from foreground to background. Besides the global discriminator $D_h$, a domain discriminator $D_{\text{ver}}$ is also required to distinguish whether the foreground and the background achieve distribution alignment to pursue more significant improvement. Specifically, we construct real and fake samples by grouping image pairs of $(I \circ M, I \circ (1 - M))$ and $(\hat{I} \circ M, \hat{I} \circ (1 - M))$, respectively. A domain encoder $E_{\text{ver}}$ extracts the feature representation of the foreground ($I \circ M$ and $\hat{I} \circ M$) and background ($I \circ (1 - M)$ and $\hat{I} \circ (1 - M)$). Similar to (3), where $\circ$ indicates the Hadamard product. The extraction process of foreground and background feature representation can be formulated as

$$\begin{cases} \text{Feat}_f = E_{\text{ver}}(I \circ M) \\ \text{Feat}_b = E_{\text{ver}}(I \circ (1 - M)) \end{cases} \quad (11)$$

$$\begin{cases} \widehat{\text{Feat}}_f = E_{\text{ver}}\left(\hat{I} \circ M\right) \\ \widehat{\text{Feat}}_b = E_{\text{ver}}\left(\hat{I} \circ (1 - M)\right) \end{cases} \quad (12)$$

where $\text{Feat}_f$ and $\text{Feat}_b$ denote the feature representations extracted from the foreground and background in the real-world image $I$, and $\widehat{\text{Feat}}_f$ and $\widehat{\text{Feat}}_b$ are extracted from the harmonized image $\hat{I}$. Due to this domain discriminator $D_{\text{ver}}$, the feature distributions of foreground and background are further aligned after the harmonization operation

$$D_{\text{ver}}(I, M) = \text{Feat}_f \cdot \text{Feat}_b \quad (13)$$

where $\cdot$ means the inner product of tow vectors. The domain verification loss $\mathcal{L}_{\text{ver}}$ can be described as

$$\mathcal{L}_{\text{ver}}\left(D_{\text{ver}}, I, \hat{I}, M\right) = \mathbb{E}_I[\max(0, 1 - D_{\text{ver}}(I, M))] + \mathbb{E}_{\hat{I}}\left[\max\left(0, 1 + D_{\text{ver}}\left(\hat{I}, M\right)\right)\right] \quad (14)$$

and

$$\mathcal{L}_{\text{ver}}(G_h, I_{\text{inter}}, M) = -\mathbb{E}_{I_{\text{inter}}}[D_{\text{ver}}(G_h(I_{\text{inter}}, M), M)]. \quad (15)$$

To sum up, the full objective of the IH module consists of three items, which can be expressed as

$$\mathcal{L}\left(D_h, D_{\text{ver}}, I, \hat{I}, M\right) = \lambda_1 \mathcal{L}_{\text{adv}}\left(D_h, I, \hat{I}\right) + \lambda_2 \mathcal{L}_{\text{ver}}\left(D_{\text{ver}}, I, \hat{I}, M\right) \quad (16)$$

$$\mathcal{L}(G_h, I, I_{\text{inter}}, M) = \lambda_1 \mathcal{L}_{\text{adv}}(G_h, I_{\text{inter}}, M) + \lambda_2 \mathcal{L}_{\text{ver}}(G_h, I_{\text{inter}}, M) + \lambda_3 \mathcal{L}_{\text{rec}}(G_h, I, I_{\text{inter}}, M) \quad (17)$$

where $\lambda_1 = \lambda_2 = 1, \lambda_3 = 100$.

In view of the lack of public remote sensing ship IH dataset, we used the general benchmark synthesized dataset iHarmony4 [44] to train the IH model. To distinguish the images of different processing stages, we named the images generated by our pipeline as aligned images. The IH module established the coordination connection between the two independent operation stages of BA submodule and FA submodule and adjusted the characteristics, such as foreground brightness representation, so that the whole aligned image looked like the captured real-world image.

## IV. Implementation

### A. Dataset Statistics

FGSCR-42 [47] is a dataset for remote sensing image fine-grained ship classification. However, the long-tail characteristics of the dataset cause the performance imbalance of the model in different categories. Considering the convenient implementation and intuitive comparison, we selected 12 categories of ships. Table II shows the detailed information of real-world images in experimental dataset.

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TABLE II
DETAILED INFORMATION OF REAL-WORLD IMAGE IN EXPERIMENTAL DATASET

| Main Category | Subordinate Category | Aligned Image | Train | Test |
|---|---|---|---|---|
| Aircraft_carrier | **Kittyhawk-class_aircraft_carrier** | √ | 34 | 34 |
| | **Kuznetsov-class_aircraft_carrier** | √ | 34 | 34 |
| | Nimitz-class_aircraft_carrier | × | 388 | 165 |
| | Midway-class_aircraft_carrier | × | 146 | 62 |
| Landing_ship | **Lzumo-class_helicopter_destroyer** | √ | 32 | 31 |
| | Whitby_island-class_dock_landing_ship | × | 195 | 83 |
| Suppoet_ship | **Sacramento-class_support_ship** | √ | 25 | 25 |
| Destroyer | **Asagiri-class_destroyer** | √ | 35 | 35 |
| | **Murasame-class_destroyer** | √ | 32 | 31 |
| | Arleigh_Burke-class_destroyer | × | 407 | 174 |
| Combat_ship | Independence-class_combat_ship | × | 148 | 62 |
| Assault_ship | Wasp-class_assault_ship | × | 318 | 135 |

### B. Implementation Details

Here, all experiments were implemented via Ubuntu 16.04 LTS operation system, NVIDIA GeForce GTX 3090 GPU with 24-GB memory, PyTorch 1.9.0, CUDA 11.1, and Python 3.8.5.

For our proposed data augmentation pipeline, we needed to train two I2I translation networks (BA submodule and FA submodule, respectively) and an IH model. We optimized two Lo-CycleGAN models for 200 epochs with batchsize 8, initial learning rate 0.0002, and 100 epochs to linearly decay learning rate to 0. In the IH process, we followed the network architecture and parameter setting of [46]. The IH model trained Adam [48] optimizer with a learning rate of 0.0002 and optimized the model with objective (16). The model was trained for 100 epochs with general synthesized dataset iHarmony4 [44] and batchsize set to 12. During the experiment, the input image size of the proposed pipeline and fine-grained classification models was $256 \times 256$.

In the remote sensing image fine-grained ship classification task, we selected eight typical fine-grained classification algorithms to ensure the adequacy of the experimental conclusions, namely, ResNet-110 [2], ResNext [3], DenseNet-121 [49], PyramidNet [50], Wide Residual Network (WRN) [23], ShuffleNet-v2 [51], EfficientNet-v2 [52], and Swin-Transformer [53]. Except for epoch, these classification algorithms had the same superparameter settings in training. Specifically, ResNet-110 was trained for 200 epochs, and other classification models were trained for 100 epochs. We used Stochastic Gradient Descent (SGD) with a momentum of 0.9 and weight decay of 0.0001. For the baseline in the fine-grained classification experiment, the training data of these models were all real-world images (see Table II). The model did not conduct data augmentation in training.

### C. Evaluation Metrics

In fine-grained classification tasks, the top-$k$ accuracy rate (AR) of each category was one of the most representative performance evaluation indicators (generally speaking, the index difficulty decreased with the increase of $k$ value). In this section, the quality performance of our RSSA-12 was evaluated by Top-1 accuracy. Generally, the Top-1 AR of the $i$th class was expressed as

$$\mathrm{AR}_i^{\mathrm{Top}-1} = \frac{\mathrm{Num}_i^c}{\mathrm{Num}_i} \tag{18}$$

where $\mathrm{Num}_i^c$ represented the correct number of each category in the test set and $\mathrm{Num}_i$ represented the number of each category in the test set.

To intuitively compare the effects of our data augmentation pipeline, we calculated the mean AR (%) of 12 classes (mean at 12) and the mean AR (%) of six classes with aligned images (mean at 6). Then, these metrics were expressed as

$$\text{mean } @6 = \frac{\sum_{i=1}^{6} \mathrm{AR}_i^{\mathrm{Top}-1}}{6}, \quad \text{mean } @12 = \frac{\sum_{i=1}^{12} \mathrm{AR}_i^{\mathrm{Top}-1}}{12}. \tag{19}$$

## V. EXPERIMENTAL RESULTS

### A. Correlation Between Aligned Images and Fine-Grained Classification Performance

This experiment's purpose is to explore the relationship between the number of images generated by our pipeline in the training set and the algorithm performance. We supplemented the simulated images and aligned images to the training set in proportion and batch, sorted them into different experimental groups, and developed experiments under the same parameter conditions and test set. Tables III and IV list the classification results of ResNet-110 when supplementing simulated images and aligned images in the training set, respectively. Tables V and VI show the classification results of ResNext when supplementing simulated images and aligned images in the training set, respectively. Note that to eliminate interference and ensure the reliability of experimental results, we did not use other data augmentation methods in this experiment.

From Tables III to VI, we demonstrated that these models reached ideal results when the supplementary aligned image and simulated image were seven times that of various images in the original training set. To reveal the changing trend of

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TABLE III

FINE-GRAINED CLASSIFICATION AR (%) OF RESNET-110 WHEN ADDING SIMULATED IMAGES WITH DIFFERENT PROPORTIONS

| *Sim-ResNet* | $AR_1$ | $AR_2$ | $AR_3$ | $AR_4$ | $AR_5$ | $AR_6$ | $AR_7$ | $AR_8$ | $AR_9$ | $AR_{10}$ | $AR_{11}$ | $AR_{12}$ | *mean* | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | @6 | @12 |
| ***K=0*** | 57.14 | 77.42 | 67.65 | 60.00 | 64.71 | 61.29 | 88.51 | 93.33 | 100.00 | 75.90 | 98.39 | 90.32 | 64.70 | 77.89 |
| ***K=2*** | 71.43 | 74.19 | 70.59 | 68.00 | 70.59 | 58.06 | 85.63 | 92.73 | 100.00 | 80.72 | 100.00 | 91.94 | 68.81 | 80.32 |
| ***K=4*** | 71.43 | 83.87 | 73.53 | 68.00 | 70.59 | 58.06 | **93.10** | **95.15** | 100.00 | 83.13 | 100.00 | 91.94 | 70.91 | 82.40 |
| ***K=6*** | 71.43 | 90.32 | 76.47 | **68.00** | **79.41** | 61.29 | 91.95 | 93.94 | 100.00 | **85.54** | 100.00 | 91.94 | 74.49 | 84.19 |
| ***K=7*** | **77.14** | **96.82** | **79.43** | 64.00 | 76.47 | **61.29** | 91.95 | 94.55 | **100.00** | 83.13 | **100.00** | **91.94** | **75.86** | **84.73** |
| ***K=8*** | 54.29 | 96.77 | 73.53 | 60.00 | 70.59 | 61.29 | 85.06 | 91.52 | 99.26 | 79.52 | 91.94 | 88.71 | 69.41 | 79.37 |

TABLE IV

FINE-GRAINED CLASSIFICATION AR (%) OF RESNET-110 WHEN ADDING ALIGNED IMAGES WITH DIFFERENT PROPORTIONS

| *Trans-ResNet* | $AR_1$ | $AR_2$ | $AR_3$ | $AR_4$ | $AR_5$ | $AR_6$ | $AR_7$ | $AR_8$ | $AR_9$ | $AR_{10}$ | $AR_{11}$ | $AR_{12}$ | *mean* | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | @6 | @12 |
| ***K=0*** | 57.14 | 77.42 | 67.65 | 60.00 | 64.71 | 61.29 | 88.51 | 93.33 | 100.00 | 75.90 | 98.39 | 90.32 | 64.70 | 77.89 |
| ***K=2*** | 71.43 | 83.87 | 76.47 | 56.00 | 73.53 | 61.29 | 89.08 | **95.76** | 100.00 | 81.93 | 96.77 | 91.94 | 70.43 | 81.51 |
| ***K=4*** | 77.14 | 83.87 | 76.47 | 72.00 | 70.59 | 58.06 | 89.66 | 92.73 | 99.26 | 75.90 | 100.00 | 88.71 | 73.02 | 82.03 |
| ***K=6*** | 71.43 | **96.77** | 79.41 | **76.00** | 85.29 | **77.06** | 91.38 | 92.73 | 100.00 | 85.54 | 100.00 | 88.71 | 81.00 | 87.03 |
| ***K=7*** | **77.14** | 90.32 | **91.18** | 72.00 | **94.12** | 74.19 | **93.68** | 93.94 | **100.00** | **89.16** | **100.00** | **91.94** | **83.16** | **88.97** |
| ***K=8*** | 71.43 | 93.55 | 76.47 | 68.00 | 73.53 | 67.74 | 91.38 | 93.33 | 100.00 | 80.72 | 100.00 | 90.32 | 75.12 | 83.87 |

TABLE V

FINE-GRAINED CLASSIFICATION AR (%) OF RESNEXT WHEN ADDING SIMULATED IMAGES WITH DIFFERENT PROPORTIONS

| *Sim-ResNext* | $AR_1$ | $AR_2$ | $AR_3$ | $AR_4$ | $AR_5$ | $AR_6$ | $AR_7$ | $AR_8$ | $AR_9$ | $AR_{10}$ | $AR_{11}$ | $AR_{12}$ | *mean* | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | @6 | @12 |
| ***K=0*** | 77.14 | 74.19 | 67.65 | 72.00 | 85.29 | 41.94 | 74.14 | 90.91 | 89.63 | 77.11 | 96.77 | 95.16 | 69.70 | 78.49 |
| ***K=2*** | 74.29 | **96.77** | 70.59 | 52.00 | 79.41 | 61.29 | 81.61 | 88.48 | 97.04 | 72.29 | 90.32 | 96.77 | 72.39 | 80.07 |
| ***K=4*** | 57.14 | 90.32 | 91.17 | **84.00** | 26.47 | **74.19** | 78.74 | 92.73 | 98.52 | 83.13 | 95.16 | 96.77 | 70.55 | 80.70 |
| ***K=6*** | **77.14** | 87.10 | 85.29 | 64.00 | 67.65 | 70.97 | 85.63 | **97.58** | 97.04 | **85.54** | **100.00** | 95.16 | 75.36 | 84.42 |
| ***K=7*** | 71.43 | 90.32 | **94.12** | 72.00 | **85.29** | 58.06 | 83.33 | 92.73 | 97.78 | 81.93 | 95.16 | **96.77** | **78.54** | **84.91** |
| ***K=8*** | 65.71 | 90.32 | 82.35 | 48.00 | 52.94 | 54.84 | **85.63** | 83.03 | **100.00** | 77.11 | 98.39 | 96.77 | 65.69 | 77.93 |

TABLE VI

FINE-GRAINED CLASSIFICATION AR (%) OF RESNEXT WHEN ADDING ALIGNED IMAGES WITH DIFFERENT PROPORTIONS

| *Trans-ResNext* | $AR_1$ | $AR_2$ | $AR_3$ | $AR_4$ | $AR_5$ | $AR_6$ | $AR_7$ | $AR_8$ | $AR_9$ | $AR_{10}$ | $AR_{11}$ | $AR_{12}$ | *mean* | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | @6 | @12 |
| ***K=0*** | 77.14 | 74.19 | 67.65 | 72.00 | 85.29 | 41.94 | 74.14 | 90.91 | 89.63 | 77.11 | 96.77 | 95.16 | 69.70 | 78.49 |
| ***K=2*** | 74.29 | 96.77 | 70.59 | 48.00 | 85.29 | 64.52 | 80.46 | 84.85 | **98.52** | 77.11 | 95.16 | 93.55 | 73.24 | 80.76 |
| ***K=4*** | 62.86 | 87.10 | **94.12** | **84.00** | 44.12 | 74.19 | 79.89 | 86.06 | 97.04 | 83.13 | 98.39 | 93.55 | 74.40 | 82.04 |
| ***K=6*** | 82.86 | 80.65 | 58.82 | 72.00 | **100.00** | 64.52 | **91.38** | **96.36** | 95.56 | **91.57** | **100.00** | 96.77 | 76.47 | 85.87 |
| ***K=7*** | **91.43** | **100.00** | 91.18 | 68.00 | 97.06 | 67.74 | 79.31 | 91.52 | 97.78 | 79.52 | 98.39 | **98.39** | **85.90** | **88.36** |
| ***K=8*** | 77.14 | 77.42 | 58.82 | 76.00 | 91.18 | 48.39 | 84.48 | 95.15 | 94.81 | 90.36 | 98.39 | 98.39 | 71.49 | 82.54 |

performance more intuitively with the proportion of supplementary images, we drew the relationship curve between them (see Fig. 8).

According to the conclusion of this experiment, we mixed these aligned images with the real-world images in an appropriate proportion. For six categories of ships with scarce real-world images, we screened and sorted 1166 high-quality aligned images generated by our pipeline, which was about seven times that of corresponding real-world ship images. We combined these high-quality aligned images as training data with real-world images (as shown in Table II) to form a mixed dataset and conducted follow-up experiments based on the mixed dataset.

### *B. Comparison With Basic Image Manipulations*

In this section, we compared the effectiveness of traditional image data augmentation methods and our proposed pipeline. To improve the integrity of the comparative experiment, we selected seven basic data manipulations: cutmix [54], cutout [23], dual cutout, random crop, random erasing [55], random horizontal flip, and anywhere-RICAP [56].

Table VII lists the results of different data augmentation methods for ResNet-110 (100 epochs and 200 epochs), ResNext, and PyramidNet, respectively. The baseline in Table VII represents the fine-grained classification performance of the three algorithms models without any data augmentation technology. In general, the traditional image data augmentation methods played a positive role in improving the model's classification performance, especially random horizontal flip and anywhere-RICAP. As listed in Table VII, we can demonstrate that our pipeline is universality superior to other basic data augmentation methods in three classification algorithms. The experimental results indicated that when our method was used together with the traditional image data augmentation methods (the combination in Table VII), the

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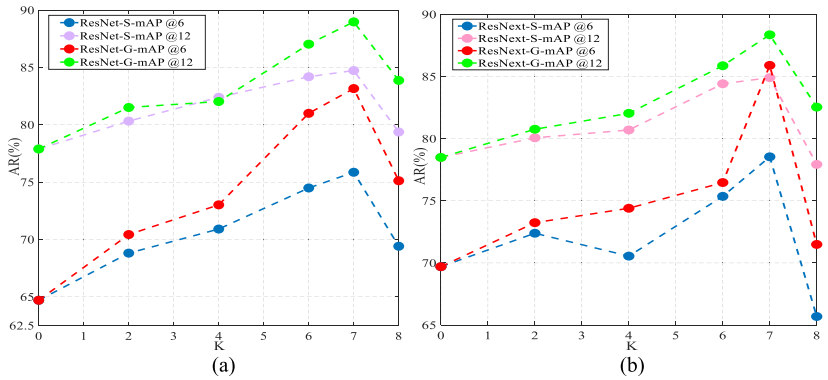



Fig. 8. Curves of classification accuracy with supplementary images with different proportions. Fine-grained classification accuracy of (a) ResNet-110 and (b) ResNext. "S" indicates supplementary simulated image and "G" indicates supplementary aligned image.

TABLE VII

AR (%) WITH OUR PIPELINE AND COMMON TRADITIONAL IMAGE DATA AUGMENTATION METHODS FOR RESNET-110 (100 AND 200 EPOCHS), RESNEXT, AND PYRAMIDNET

| Method | ResNet-110_100 Epochs | | ResNet-110_200 Epochs | | ResNext_100 Epochs | | PyramidNet_100 Epochs | |
|---|---|---|---|---|---|---|---|---|
| | *mean* @6 | *mean* @12 | *mean* @6 | *mean* @12 | *mean* @6 | *mean* @12 | *mean* @6 | *mean* @12 |
| Baseline | 62.59 ±1.32 | 76.88 ±0.76 | 67.50 ±2.80 | 79.44 ±1.55 | 74.44 ±1.82 | 83.95 ±0.75 | 75.57 ±0.85 | 84.48 ±0.34 |
| + cutmix [54] | 62.65 | 76.99 | 70.27 | 80.90 | 80.16 | 84.86 | 77.97 | 86.06 |
| + cutout [23] | 62.54 | 76.68 | 67.50 | 79.30 | 71.01 | 81.06 | 79.22 | 86.48 |
| + dual cutout | 62.41 | 77.41 | 69.72 | 80.19 | 79.20 | 85.32 | 78.92 | 86.47 |
| + random crop | 65.41 | 78.87 | 74.07 | 82.83 | 75.21 | 83.79 | 76.88 | 85.98 |
| + random erasing [55] | 67.46 | 80.00 | 67.75 | 79.62 | 76.41 | 85.61 | 81.38 | 87.82 |
| + random horizontal flip | 66.35 | 78.95 | 73.13 | 82.02 | 77.91 | 86.28 | 83.19 | 89.23 |
| + anywhere-RICAP [56] | 68.30 | 79.93 | 75.00 | 83.69 | 77.83 | 85.78 | 82.62 | 88.46 |
| Our Method | 73.03 ±0.52 | 82.98 ±0.11 | 82.87 ±0.30 | 89.07 ±0.10 | 84.10 ±0.94 | 88.93 ±0.10 | 83.49 ±2.09 | 89.61 ±1.68 |
| Combination | **75.07** | **85.01** | **85.00** | **90.95** | **86.04** | **90.49** | **87.42** | **92.00** |
| **Gain** | **+12.48** | **+8.13** | **+17.5** | **+11.51** | **+11.6** | **+6.54** | **+11.85** | **+7.52** |

fine-grained classification accuracy would be improved. The mean at 6 of combination on these algorithms was 12.48%, 17.5%, 11.6%, and 11.85% higher than the baseline, respectively. The mean at 12 of combination on these algorithms was 8.13%, 11.51%, 6.54%, and 7.52% higher than the baseline, respectively.

The long-tail distribution between classes in the real-world dataset is one of the primary reasons for performance imbalance. In Table VII, the difference between mean at 6 and at 12 of the classification algorithm is positively related to its performance imbalance. Taking the ResNext in Table VII as an example, the difference between mean at 6 and at 12 is reduced from 9.51% (baseline) to 4.45% (combination). This reduction reveals that the imbalance in model performance has been improved by supplementing our aligned images. In other words, our data augmentation pipeline exhibits concrete advantages in alleviating the model performance imbalance caused by the data long-tail distribution.

### C. Comparison With Other Translation Methods

In addition to the common traditional image data augmentation methods, we compared other translation methods based on DL technology. CycleGAN [25] and Sim2RealNet [32] were the representative works of global-aware feature learning and NST, respectively. SemI2I [57] and CUT [39] were the recent image translation works of remote sensing image processing field and contrastive learning, respectively.

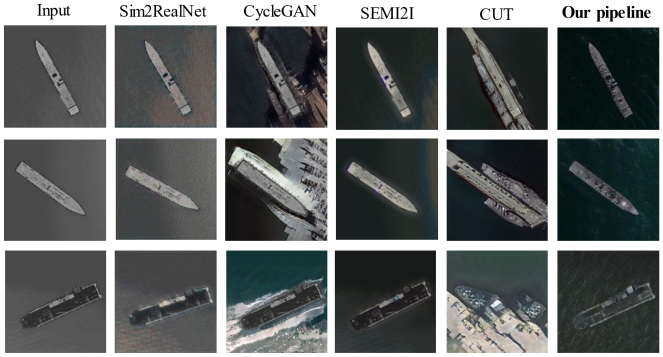



Fig. 9. Visual appearance comparison between our aligned image and the results generated by other representative methods.

In Fig. 8, we compared our aligned image with the results generated by these models.

As Fig. 9 shows, Sim2RealNet and SEMI2I could not completely change the entire appearance style of the image. Moreover, since the ocean background accounted for a large proportion of the image, the deep model tended to extract the style feature distribution of the background region, rather than the visual expression feature of the ship target. We used CycleGAN and CUT for global-aware feature alignment, and the image background region achieved positive visual alignment. However, due to the feature interference of another region, the global-aware generated image would be deformed

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TABLE VIII
COMPARISON RESULTS WITH EXISTING TRANSLATION METHODS ON REPRESENTATIVE CLASSIFICATION ALGORITHMS

| Models | Methods | $AR_1$ | $AR_2$ | $AR_3$ | $AR_4$ | $AR_5$ | $AR_6$ | $AR_7$ | $AR_8$ | $AR_9$ | $AR_{10}$ | $AR_{11}$ | $AR_{12}$ | *mean* @6 | *mean* @12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ResNet-110 (200 epochs) | Baseline | 74.29 | 74.19 | 67.65 | 56.00 | 61.76 | 58.06 | 85.63 | 92.73 | 100.00 | 77.11 | 100.00 | 90.32 | 65.33 | 78.15 |
| | Only-Sim | 80.00 | 80.30 | 76.50 | 76.00 | 72.40 | 64.50 | **94.30** | 92.70 | 100.00 | 80.70 | 100.00 | 88.70 | 74.95 | 83.84 |
| | CycleGAN | 74.50 | 68.20 | 56.80 | 62.10 | 65.00 | 48.00 | 88.30 | 93.30 | 97.10 | 80.50 | 98.30 | 88.40 | 62.43 | 76.71 |
| | Sim2RealNet | **82.86** | 87.10 | 84.71 | **78.00** | 82.35 | 74.66 | 87.36 | 94.55 | **100.00** | 84.34 | 100.00 | 91.94 | 81.61 | 87.32 |
| | SEMI2I | 80.00 | 80.65 | 79.41 | 75.00 | 70.59 | 74.66 | 90.80 | 92.73 | 100.00 | 84.34 | 98.39 | 87.10 | 76.72 | 84.47 |
| | CUT | 51.43 | 83.87 | 67.65 | 60.00 | 64.71 | 45.16 | 82.18 | 89.70 | 100.00 | 83.13 | 96.77 | 90.32 | 62.14 | 76.24 |
| | Our pipeline | 81.10 | **90.30** | **94.10** | 72.00 | **100.00** | **77.40** | 88.50 | **98.20** | 97.00 | **91.60** | **100.00** | **93.50** | **85.82** | **90.31** |
| ShuffleNet-v2 (100 epochs) | Baseline | 78.57 | 95.83 | 78.57 | 65.00 | 91.67 | 68.00 | 94.25 | 99.39 | 100.00 | 91.57 | 100.00 | 98.39 | 79.61 | 88.44 |
| | Only-Sim | 75.00 | 100.00 | 75.00 | 65.00 | 87.50 | 76.00 | 93.10 | 95.76 | 100.00 | 92.77 | 100.00 | 95.16 | 79.75 | 87.94 |
| | CycleGAN | 71.43 | 87.10 | 85.29 | 68.00 | 82.50 | 74.19 | 94.25 | 98.79 | 100.00 | 90.36 | 98.39 | 91.94 | 78.09 | 86.85 |
| | Sim2RealNet | 82.86 | 90.32 | 82.35 | 88.00 | **100.00** | 67.74 | 86.78 | 95.76 | 99.26 | 90.36 | 96.77 | 88.71 | 85.21 | 89.08 |
| | SEMI2I | 68.57 | 96.77 | **88.24** | 68.00 | 97.06 | **80.65** | 92.53 | **98.79** | 100.00 | 92.77 | 100.00 | 95.16 | 83.22 | 89.88 |
| | CUT | 60.00 | 100.00 | 88.24 | 64.00 | 82.35 | 67.74 | 90.80 | 98.79 | 99.26 | 91.57 | 100.00 | 96.77 | 77.06 | 86.63 |
| | Our pipeline | **82.86** | **100.00** | 85.29 | **88.00** | 94.12 | 74.19 | **95.98** | 97.58 | **100.00** | **95.18** | **100.00** | **98.39** | **87.41** | **92.63** |
| SwinT (100 epochs) | Baseline | **89.29** | 91.67 | 100.00 | 75.00 | 95.83 | 76.00 | 85.06 | 94.55 | 95.56 | 96.39 | 100.00 | 98.39 | 87.97 | 91.48 |
| | Only-Sim | 85.71 | 100.00 | 96.43 | 75.00 | 91.67 | 80.00 | 94.25 | 98.18 | 97.04 | **97.59** | 100.00 | 100.00 | 88.14 | 92.99 |
| | CycleGAN | 82.86 | 77.42 | 85.29 | 92.00 | 100.00 | 80.65 | 93.68 | 99.39 | 95.56 | 93.98 | 98.39 | 96.77 | 86.37 | 91.33 |
| | Sim2RealNet | 82.86 | **100.00** | 88.24 | 92.00 | 100.00 | 83.87 | 91.95 | **100.00** | 98.52 | 93.98 | 95.16 | 98.39 | 91.16 | 93.75 |
| | SEMI2I | 82.86 | 100.00 | 100.00 | 92.00 | 100.00 | 67.74 | 92.53 | 100.00 | 99.26 | 91.57 | 100.00 | 100.00 | 90.43 | 93.83 |
| | CUT | 85.71 | 96.77 | 82.35 | 80.00 | 79.41 | 83.87 | 85.63 | 98.79 | 98.52 | 97.59 | 98.39 | 93.55 | 84.69 | 90.05 |
| | Our pipeline | 85.71 | 80.65 | **100.00** | **96.00** | **100.00** | **100.00** | **96.55** | 99.39 | **100.00** | 93.98 | **100.00** | **100.00** | **93.73** | **96.02** |

or distorted. We used the same number of images generated by these methods to supplement the real-world image training set and implemented comparative experiments under the same parameters and test set. The experiment was implemented on ResNet-110, ShuffleNet-v2, and Swin Transformer (see Table VIII).

In Table VIII, due to the chaotic expression of the image generated by the CycleGAN and CUT, the fine-grained classification accuracy of three representative algorithms was weakened to varying degrees. Also, other data augmentation methods have played a positive role in varying degrees. For the three representative classification algorithms, our pipeline is nearly 3% (mean at 6) and 2.94% (mean at 12) higher than the second-best method (Sim2RealNet [32]) on average.

Neural network interpretability research is devoted to interpret and explicit the working principle and decision-making process of deep networks. Class activation mapping (CAM) [58], [59] is one of the representative achievements, which shows the network model high-activation area on the input image with the heat map. Generally, the coverage ratio of the high-activation area to the ship target demonstrates the fine-grained feature representation capacity of the model. For the classification model trained on the images generated by different methods, we use the Grad-CAM [60] to visualize the output of the deep network's last convolution layer, as shown in Fig. 10. The visualization results indicate that the model trained by our pipeline generated image has a larger high-activation area on the ship target, which demonstrates that the model has a stronger ability to extract and represent the fine-grained features of the ship target. Especially, the high-activation area of the model trained on the CycleGAN and CUT generated images has an inapparent coincidence with the ship target, which confirms the inferior performance of the relevant classification models from the visual interpretation perspective.

### D. Ablation Study

We conducted extensive ablation studies to demonstrate the necessity of each stage in our proposed method. We decomposed and combined the module (submodule) in our pipeline and evaluated the effectiveness of these from the perspective of visual expression and algorithm performance. Fig. 11 shows the visual expression of the generated image by combining different modules.

Fig. 11 shows the generated image examples of each critical stage. With the iteration of training epochs in the BA submodule, the style expression features of the sea surface area of the simulated image tended to the real-world domain. Note that in this process, it maintained other dimensional features of the background region, such as the brightness change pattern of the sea surface, the texture shape of waves, and so on. Then, we solidified the representation of the background to ensure its invariance. With the training iteration in the FA process, the style representation of ship target was transferred from the simulation domain to the real-world domain. Considering that the two alignment processes were independent of each other, the visual effects of foreground and background are inharmonious. Specifically, the luminance of the foreground was higher than that of the background, similar to "overexposure," as shown in the fifth column from left to right (see Fig. 11). Also note that with the superposition of the implementation results of each process, the detailed characteristics of ship targets were prominent, and the appearance representation between regions was more coordinated. The vividness of image visual presentation enriches the availability of multidimensional information and contributes to improve the accuracy of ship fine-grained classification.

Since our work was a progressive process of multistage feature alignment, to analyze the effect of each module in detail, we decomposed and reorganized each module of our

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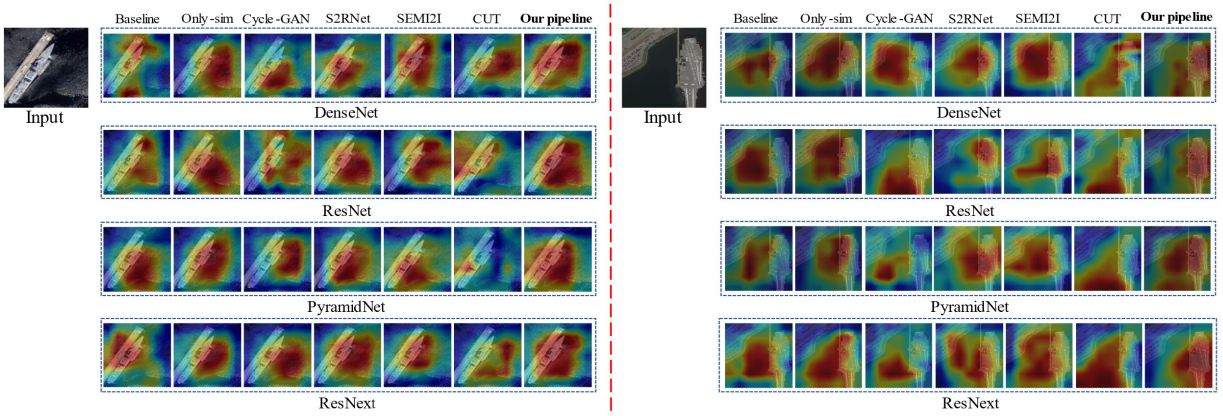



Fig. 10. Visualization results of representative classification algorithms with different remote sensing image training. "S2RNet" in this figure is the abbreviation of "Sim2RealNet."

TABLE IX
RESULTS OF OUR ABLATION STUDY

| *BA* | *FA* | *IH* | ResNet-110_200 Epochs | | ResNext_100 Epochs | | PyramidNet_100 Epochs | | EffiNet-v2_100 Epochs | | SwinT_100 Epochs | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | *mean* @6 | *mean* @12 | *mean* @6 | *mean* @12 | *mean* @6 | *mean* @12 | *mean* @6 | *mean* @12 | *mean* @6 | *mean* @12 |
| | Baseline | | $67.50_{\pm 2.80}$ | $79.44_{\pm 1.55}$ | $74.44_{\pm 1.82}$ | $83.95_{\pm 0.75}$ | $75.57_{\pm 0.85}$ | $84.48_{\pm 0.34}$ | $83.68_{\pm 1.22}$ | $89.10_{\pm 0.88}$ | $87.32_{\pm 1.29}$ | $91.53_{\pm 0.65}$ |
| | All-Sim | | 75.86 | 84.73 | 78.54 | 84.91 | 77.37 | 85.33 | 86.16 | 90.56 | 88.14 | 92.99 |
| √ | | | 77.90 | 87.00 | 75.17 | 85.55 | 80.53 | 87.22 | 87.23 | 90.70 | 87.37 | 92.24 |
| √ | √ | | 81.69 | 89.13 | 80.91 | 89.02 | 83.81 | 89.30 | 88.90 | 92.31 | 91.48 | 94.48 |
| √ | √ | √ | **85.00** | **90.95** | **84.41** | **90.14** | **85.57** | **91.28** | **90.65** | **93.69** | **93.73** | **96.02** |
| | Gain-BA | | +2.04 | **+2.27** | -3.37 | +0.64 | +3.16 | +1.89 | +1.07 | +0.14 | -0.77 | -0.75 |
| | Gain-FA | | **+3.79** | +2.13 | **+5.74** | **+3.47** | **+3.28** | **+2.08** | +1.67 | **+1.61** | **+4.11** | **+2.24** |
| | Gain-IH | | +3.31 | +1.82 | +3.50 | +1.12 | +1.76 | +1.98 | **+1.75** | +1.38 | +2.25 | +1.54 |

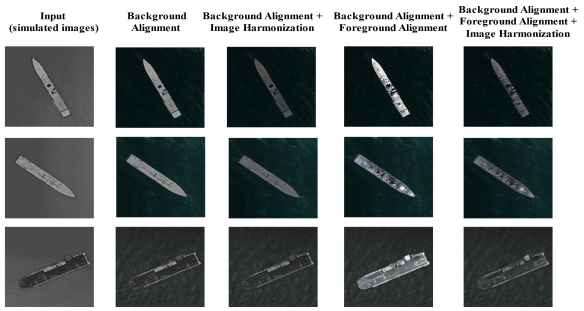



Fig. 11. Visual expression of the generated image by the combination of different modules.

proposed pipeline. Table IX summarizes the performance of each classification algorithm under different process combinations. The BA submodule optimized the classification performance of the three models by 0.43% (mean at 6) and 0.84% (mean at 12) on average. Similarly, the FA submodule was 3.72% (mean at 6) and 2.31% (mean at 12), and the IH module was 2.51% (mean at 6) and 1.57% (mean at 12). Each module or submodule, especially FA submodule, improved the performance of the model. The excellent effect of FA submodule benefited from the vivid expression of the detailed features of ship targets during this process, which was conducive to acquire discriminant features by the deep model, and then enhanced the comprehensiveness and precision of image information representation. The positive advances of each module (submodule) revealed that the background, foreground, and the global-expression would affect the interpretation of image content and the fine-grained classification ability.

### *E. Benchmark Results*

In this section, we organize and mix the aligned images with a few real-world images, add high-precision category labels, and publish a mixed dataset for remote sensing ship fine-grained classification task. Based on the analysis of the experimental results shown in Fig. 8, we set the number of aligned images in a reasonable range to ensure the comprehensive index of the dataset. The dataset was named RSSA-12, meaning a dataset for Remote Sensing fine-grained Ship classification in Aligned images with 12 categories. Six of these categories (Kittyhawk-class_aircraft_carrier, Kuznetsov-class_aircraft_carrier, Lzumo-class_helicopter_destroyer, Sacramento-class_support_ship, Asagiri-class_destroyer, and Murasame-class_destroyer) are composed of our aligned images and real-world images. As the real-world data of these six types of ships are very scarce, we collected some real-world images from FGSCR-42 and added them to our dataset after image processing. The remaining six categories are all real-world images. These real-world images are filtered and sorted from FGSCR-42 and have undergone some image processing operations. Some sample images with annotation information in RSSA-12 are shown in Fig. 12.

We compared four representative ship datasets in remote sensing images. DOTA [61], NWPU VHR-10 [62], and DSCR [63] contained fewer ship types, which were not enough for fine-grained research. FGSCR-42 [47] was limited by the long-tail problem and lacked diversity in describing the target characteristics with scarce real-world images. Different from these above datasets, our RSSA-12 included multiple

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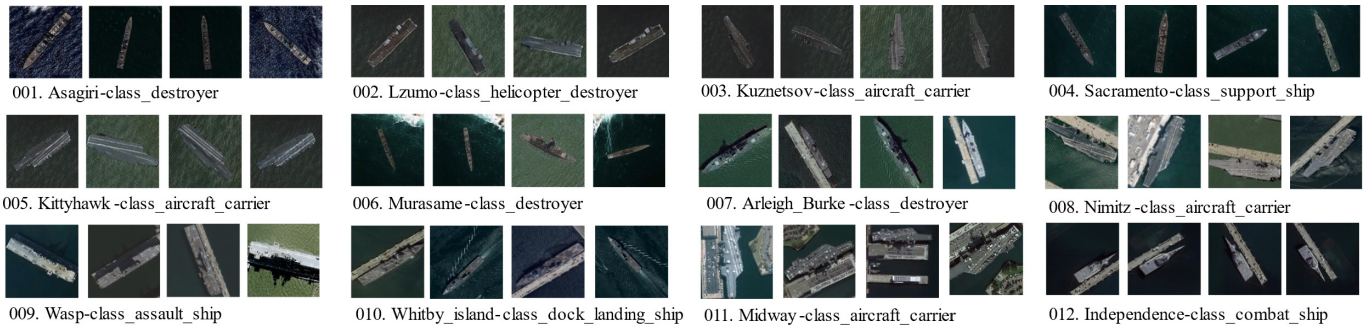



Fig. 12. Sample images of our RSSA-12 dataset.

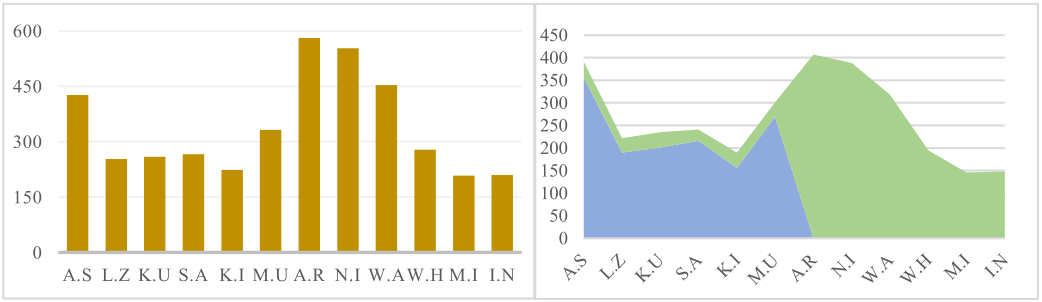



Fig. 13. Number of ships in each category in our dataset. (a) Total number of various categories of ship images in the dataset, including the real-world image and the aligned images generated by our pipeline. (b) Proportion of real-world images and aligned images in the training set of RSSA-12. The blue region represents the images generated by our pipeline, and the green region represents the number of real-world images.

TABLE X

FINE-GRAINED CLASSIFICATION RESULTS OF GENERAL CLASSIFICATION NETWORKS [TOP-1 AR (%)]

| Type | Model | *mean* @6 | *mean* @12 |
|---|---|---|---|
| General CNN Classification Network | DenseNet-121 [48] | 81.32 | 87.19 |
| | ResNext [3] | 84.41 | 90.14 |
| | ResNet-110 [2] | 85.00 | 90.95 |
| | PyramidNet [49] | 85.57 | 91.28 |
| | ShuffleNet-v2 [23] | 87.41 | 92.63 |
| | WRN [50] | 90.25 | 93.92 |
| | EfficientNet-v2 [51] | 90.65 | 93.69 |
| Transformer | Swin-Transformer [52] | 93.73 | 96.02 |

TABLE XI

EXPERIMENTAL RESULTS OF AR (%) FOR EACH CATEGORY ON WRN

| **Category** | **Accuracy** |
|---|---|
| 001. Asagiri-class_destroyer | 82.86 |
| 002. Lzumo-class_helicopter_destroyer | 100.00 |
| 003. Kuznetsov-class_aircraft_carrier | 97.06 |
| 004. Sacramento-class_support_ship | 100.00 |
| 005. KittyHawk-class_aircraft_carrier | 97.06 |
| 006. Murasame-class_destroyer | 64.52 |
| 007. Arleigh_Burke-class_destroyer | 94.25 |
| 008. Nimitz-class_aircraft_carrier | 99.39 |
| 009. Wasp-class_helicopter_destroyer | 100.00 |
| 010. Whitby_Island-class_dock_landing_ship | 95.18 |
| 011. Midway-class_aircraft_carrier | 100.00 |
| 012. Independence-class_combat_ship | 96.77 |

categories of ships, and the aligned images were supplemented to alleviate the scarcity of real-world images. The properties of our RSSA-12 are given in the following.

Fig. 13 shows the number distribution of images in each category of the dataset, and Fig. 13(b) shows that the real-world image had a noticeable long-tail effect, which was unfavorable to the performance of fine-grained classification. Benefiting from the aligned images, the number of various ship images in RSSA-12 training set is relatively balanced, which positively alleviates the long-tail problem in the real-world dataset. To pledge the comprehensiveness of the algorithm and the experimental integrity, we selected seven general-purpose CNN classification networks (ResNet-110, ResNext, DenseNet-121, PyramidNet, ShuffleNet-v2, WRN, and EfficientNet-v2) and Swin-Transformer to build a benchmark on our RSSA-12. Table X lists the performance of each algorithm.

We listed the AR (%) of each category when using the WRN for remote sensing image fine-grained ship classification on our dataset (see Table XI). Analyzing the accuracy of each category allowed us to obtain the characteristics of datasets and algorithms.

Considering the particularity of the Lzumo-class_helicopter_destroyer and Wasp-class_helicopter_destroyer, their contour and appearance characteristics were close to the aircraft carrier. From Table XI, we concluded that the accuracy of the exact destroyer was lower than that of the aircraft carrier, that is, the classification accuracy of ships with large spatial size and distinctive shape characteristics was

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relatively satisfying. Similar conclusions could be profiled by analyzing the classification accuracy of other algorithms. It might be that small spatial scale targets contained fewer discrimination features and large intraclass differences, which limited the algorithm to extract detailed features.

## VI. Conclusion

Here, we proposed a novel data augmentation pipeline with simulated image and LoPIT. To tackle the disadvantages of image translation based on image-level feature learning, we proposed Lo-CycleGAN to achieve the mapping based on local representation learning and progressively complete the remote sensing image global cross-domain translation. On this basis, we used an IH model to improve image feature appearance and generated sufficient remote sensing ship images with precise representation, photorealistic style, and harmonious features. Meanwhile, we built up a mixed dataset including real-world images and our synthetic images for remote sensing image fine-grained ship classification. Extensive experiment results demonstrated that our progressive pipeline performed positively against the state-of-the-art data augmentation method in terms of the performance on the remote sensing fine-grained ship classification task.

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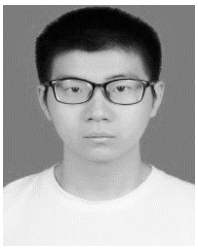
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